

There are a total of six problems. You have to solve all the problems.

**Problem 1 (CO1): Finite Automata (10 points)**

Let  $\Sigma = \{0, 1\}$ . Consider the following languages over  $\Sigma$ .

$$L_1 = \{001, 011, 101, 111\}$$

$$L_2 = \{w : w \text{ contains exactly two 0s between every successive 1s}\}$$

$$L_3 = L_1^* \cap L_2$$

$$L_4 = \{w : w \text{ contains } x \text{ as a substring where } x \in L_3 \text{ and } |x| > 0\}$$

Now solve the following problems.

- Give** the state diagram of a DFA that recognizes  $L_1^*$ . [Note:  $L_1$  is a finite language consisting of four strings] (3 points)
- Give** the state diagram of a DFA that recognizes  $L_2$ . (3 points)
- Write** all strings of length at most ten in  $L_3$ . (1 point)
- What** is the minimum number of states required to construct an NFA for  $L_3$ . (1 point)
- Give** the state diagram of a DFA that recognizes  $L_4$ . [Recall:  $|x|$  denotes the length of the string  $x$ ] (2 points)

**Problem 2 (CO2): Converting Regular Expressions to NFA (10 points)**

**Convert** the following regular expression over  $\Sigma = \{a, b\}$  into an equivalent NFA. Note that  $R_1 + R_2$  is the same as  $R_1 \cup R_2$ .

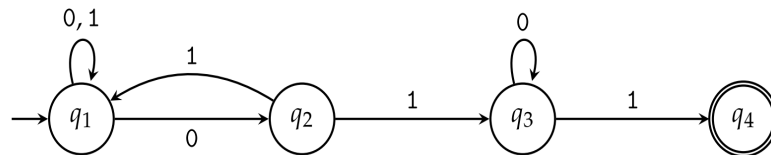
$$b((aa + b)^* + a)^*aa$$

**Problem 3 (CO1): Regular Languages (5 points)**

Answer the following questions.

- For  $\Sigma = \{0, 1, 2\}$ , **what** is the total number of transitions in a DFA with 7 states? (1 point)
- Write** the smallest string generated by the following regular expression:  $bba^*(ab + (aa)^*a) + aa(aa + bab)(a + \epsilon)$ . (1 point)
- Write** the regular expression for  $\phi.R$ , where  $R$  is a regular expression and  $\phi$  denotes the empty language. (1 point)
- Write** whether the regular expressions  $(0 + \epsilon)(1 + 10)^*$  and  $1^*(011^*)^*(0 + \epsilon)$  are equivalent or not. (1 point)

Consider the following NFA:



Now consider the following the regular languages:

$$L_1 = \{w \in \{0, 1\}^* : w \text{ contains at least two 1s}\}$$

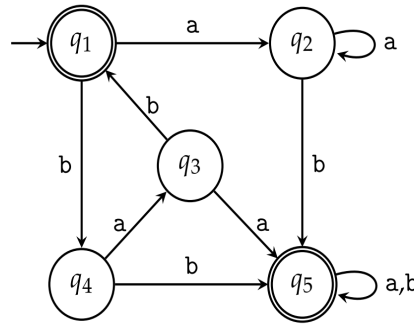
$$L_2 = \{w \in \{0, 1\}^* : w \text{ contains the substring } 011\}$$

$$L_3 = \{w \in \{0, 1\}^* : w \text{ ends with } 1\}$$

- Write** the correct option below that specifies the languages recognized by the given NFA. (1 point)
- (i) Only  $L_2$    (ii) Only  $L_3$    (iii)  $L_2$  and  $L_3$    (iv)  $L_1, L_2$  and  $L_3$    (v) None of  $L_1, L_2$  and  $L_3$

Problem 4 (CO2): Converting FA to Regular Expressions (10 points)

**Convert** the following finite automata into an equivalent regular expression using the state elimination method. You must eliminate the states in the following order:  $q_2, q_3, q_4, q_1, q_5$ . Show each step of the process.



Problem 5 (CO1): Regular Expressions (10 points)

Let  $\Sigma = \{0, 1\}$ . Consider the following languages over  $\Sigma$ .

$$L_1 = \{w : w \text{ starts and ends with } 0\}$$

$$L_2 = \{w : |w| \% 3 = 0\}$$

$$L_3 = \{w = 0^n 1^m : n \text{ is odd and } m \text{ is even}\}$$

$$L_4 = \{w : w \text{ contains even number of } 10 \text{ as a substring}\}$$

$$L_5 = L_1 \cap L_2$$

Now solve the following problems.

- Give** a regular expression that generates  $L_1$ . (2 points)
- Give** a regular expression that generates  $\overline{L_2}$ . [Recall:  $\overline{L}$  denotes the complement of the language  $L$  i.e.,  $\overline{L} = \Sigma^* - L$ ] (2 points)
- Give** a regular expression that generates  $L_3$ . (2 points)
- Give** a regular expression that generates  $L_4$ . (2 points)
- Give** a regular expression that generates  $L_5$ . (2 points)

Problem 6 (CO2): Converting NFA to DFA (5 points)

Write whether the following statements are true or false. In addition, write the correct statements for the false statements.

- When converting an NFA to an equivalent DFA using the subset construction method, the resulting DFA can be exponentially larger than the original NFA. (1 point)
- Using the subset construction method it can be proved that every language that is recognized by a DFA can also be recognized by a NFA. (1 point)

Consider the following NFA:



Now answer the following questions.

- Write** the  $\epsilon$ -closure of state  $q_1$  in the given NFA. (1 point)
- Write** the subset of NFA states that will form the starting state of the equivalent DFA. (1 point)
- What** is  $\delta(\{q_2, q_3\}, 0)$  in the given NFA? [Recall:  $\delta(\{q\}, a)$  outputs the set of states in which the NFA transitions when it is in state  $q$  and reads the symbol  $a$ ] (1 point)